

Homework 1 – Theory Part

Deep Learning – Universidad del Pacífico

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Due date: TBD

Instructions

- This part covers topics from **Lectures 1, 2 and 3**, complemented with exercises adapted from the Caltech *Learning From Data* Problem Set 4 (Yaser S. Abu-Mostafa).
- There are **6 problems** worth **100 points** in total.
- Justify every step. A correct final answer without derivation does not receive full credit.
- You may submit handwritten (clear scan/photo) or typeset in LaTeX. LaTeX is preferred.
- For network-design problems (3 and 4) you may include a diagram (TikZ, hand-drawn scan, or equivalent).

#	Topic	Points
1	Manual backpropagation of a 2-layer network	21
2	Activation functions and the vanishing gradient	16
3	Building OR with ReLU (Caltech)	12
4	Building XOR with ReLU (Caltech)	19
5	Weight initialization (Xavier and He)	16
6	The Adam optimizer: bias correction	16
Total		100

Problem 1. Manual backpropagation of a 2-layer network (21 pts)

Consider a fully connected network with the following architecture:

- Input: $\mathbf{x} \in \mathbb{R}^d$.
- Hidden layer: $\mathbf{z}^{(1)} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}$, with $W^{(1)} \in \mathbb{R}^{H \times d}$, $\mathbf{b}^{(1)} \in \mathbb{R}^H$, and activation $\mathbf{h} = \sigma(\mathbf{z}^{(1)})$ where σ is the sigmoid applied element-wise.
- Output layer: $z^{(2)} = \mathbf{w}^{(2)\top} \mathbf{h} + b^{(2)}$, with $\mathbf{w}^{(2)} \in \mathbb{R}^H$, $b^{(2)} \in \mathbb{R}$, and prediction $\hat{y} = \sigma(z^{(2)})$.
- Loss: *binary cross-entropy* $\mathcal{L} = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$, with $y \in \{0, 1\}$.

(a) (5 pts) Show that for this combination (sigmoid + BCE):

$$\frac{\partial \mathcal{L}}{\partial z^{(2)}} = \hat{y} - y.$$

- (b) (7 pts) Compute the gradients $\frac{\partial \mathcal{L}}{\partial \mathbf{w}^{(2)}}$ and $\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(2)}}$. State the shape of each tensor.
- (c) (7 pts) Compute the gradients $\frac{\partial \mathcal{L}}{\partial W^{(1)}}$ and $\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}}$ via the chain rule. Identify the factor $\sigma'(\mathbf{z}^{(1)})$ and express it as $\mathbf{h} \odot (1 - \mathbf{h})$.
- (d) (2 pts) Argue why a *single* backward pass is enough to compute every gradient, and why a finite-difference estimate would be $\mathcal{O}(P)$ times more expensive, where P is the total number of parameters.

Problem 2. Activation functions and the vanishing gradient (16 pts)

(a) (5 pts) Compute the derivatives of the following activations, expressing the result *in terms of the activation itself* whenever possible:

- $\sigma(z) = \frac{1}{1 + e^{-z}}$
- $\tanh(z)$
- $\text{ReLU}(z) = \max(0, z)$

(b) (3 pts) Show that $\sigma'(z)$ attains its maximum at $z = 0$ with value $1/4$. Conclude that in a deep **sigmoid** network the magnitude of the gradient reaching the first layer is upper bounded by

$$\|\nabla_{W^{(1)}} \mathcal{L}\| \leq C \cdot (1/4)^L,$$

for some $C > 0$ that depends on the remaining weights, where L is the number of hidden layers. Briefly discuss the practical implication.

(c) (3 pts) For $\tanh$ the analogous factor is 1 (at $z = 0$). **Why**, in practice, does $\tanh$ still fail to fully solve the vanishing gradient problem in very deep networks?

(d) (5 pts) ReLU avoids the vanishing gradient but introduces the *Dying ReLU* problem.

- (i) Formally define when a ReLU unit is *dead*.
- (ii) If a unit is dead over the entire dataset, **what is its gradient** with respect to $W^{(1)}$? What does this imply for the weight update?
- (iii) Mention *two* standard strategies (from class or from the Caltech problem set 4) to mitigate this problem.

Problem 3. Building OR with ReLU (12 pts)

Note: Adapted from Caltech Problem Set 4, Section 1.D.

Design a **fully connected ReLU network** that implements the OR function on binary inputs $x_1, x_2 \in \{0, 1\}$. The output $f(x_1, x_2)$ must satisfy:

$$\begin{aligned} \text{OR}(0, 0) &= 0, & \text{OR}(0, 1) &\geq 1, \\ \text{OR}(1, 0) &\geq 1, & \text{OR}(1, 1) &\geq 1. \end{aligned}$$

- (a) (7 pts) Provide the exact weights and biases. Use the **minimum number of hidden units** possible (justify the minimum). You may draw the network or write the matrices/vectors.
- (b) (5 pts) Verify your network on the truth table by explicitly evaluating it on the four inputs and showing the per-layer computation.

Problem 4. Building XOR with ReLU (19 pts)

Note: Adapted from Caltech Problem Set 4, Section 1.E.

Design a **fully connected ReLU network** that implements the XOR function on binary inputs:

$$\begin{aligned} \text{XOR}(0, 0) &= 0, & \text{XOR}(0, 1) &\geq 1, \\ \text{XOR}(1, 0) &\geq 1, & \text{XOR}(1, 1) &= 0. \end{aligned}$$

- (a) (5 pts) Prove formally that **no** network *without hidden layers* (i.e., a single linear layer with a ReLU at the end) can implement XOR. *Note: Hint: a function of the form $\text{ReLU}(\mathbf{w}^\top \mathbf{x} + b)$ is convex in $\mathbf{x}$. Is XOR convex?*
- (b) (5 pts) State the **minimum number of hidden layers** that allows a ReLU network to implement XOR, and justify.
- (c) (7 pts) Build an explicit minimal network: specify weights, biases, and architecture.
- (d) (2 pts) Verify your construction on the four inputs by showing the computation.

Problem 5. Weight initialization: Xavier and He (16 pts)

Consider a linear layer $\mathbf{z} = W\mathbf{x}$ with $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$, and assume the inputs x_i are i.i.d. with zero mean and variance σ_x^2 , and the weights W_{ij} are i.i.d. with zero mean, variance σ_W^2 , and **independent of $\mathbf{x}$** .

- (a) (5 pts) Show that the variance of each component z_k satisfies

$$\text{Var}(z_k) = n_{\text{in}} \sigma_W^2 \sigma_x^2.$$

Note: Recall that $\text{Var}(\sum_i A_i) = \sum_i \text{Var}(A_i)$ when the A_i are independent.

- (b) (3 pts) For $\text{Var}(z_k) = \text{Var}(x_i)$ to hold (i.e., for the signal to neither blow up nor vanish through the layer) it suffices to set $\sigma_W^2 = 1/n_{\text{in}}$. If we additionally want to preserve variance *in the backward pass*, we obtain $\sigma_W^2 = 1/n_{\text{out}}$. The **Xavier/Glorot** initialization averages both conditions:

$$\sigma_W^2 = \frac{2}{n_{\text{in}} + n_{\text{out}}}.$$

In 3-5 lines, discuss **why** this compromise makes sense and which activation functions are its *ideal case*.

- (c) (5 pts) For **ReLU** the derivation changes. Assuming $\mathbf{x}$ comes from a previous ReLU layer (so on average half of the components are zero) and that the pre-activation is symmetric around zero, show that preserving variance requires

$$\sigma_W^2 = \frac{2}{n_{\text{in}}} \quad (\text{He et al., 2015}).$$

Note: Key fact: $\mathbb{E}[\text{ReLU}(z)^2] = \frac{1}{2} \text{Var}(z)$ when $z \sim \mathcal{N}(0, \text{Var}(z))$.

(d) (3 pts) If you used **Xavier instead of He** in a deep ReLU network, what would happen to the variance of the activations as depth increases? Reason by computing the per-layer compression factor.

Problem 6. The Adam optimizer: bias correction (16 pts)

Adam combines momentum and RMSProp with a *bias correction* step. The updates are:

$$\begin{aligned}\mathbf{m}_t &= \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t, \\ \mathbf{v}_t &= \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2, \\ \hat{\mathbf{m}}_t &= \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}, \\ \boldsymbol{\theta}_t &= \boldsymbol{\theta}_{t-1} - \alpha \cdot \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon},\end{aligned}$$

with $\mathbf{m}_0 = \mathbf{v}_0 = \mathbf{0}$ and $\beta_1, \beta_2 \in [0, 1)$.

(a) (5 pts) Assume gradients are stationary, $\mathbb{E}[\mathbf{g}_t] = \mathbf{g}$ for every t . Prove by induction that

$$\mathbb{E}[\mathbf{m}_t] = (1 - \beta_1^t) \mathbf{g}.$$

Conclude that **without the correction** $\hat{\mathbf{m}}_t = \mathbf{m}_t / (1 - \beta_1^t)$ the first iterations are biased toward zero.

(b) (5 pts) Analogously, assuming $\mathbb{E}[\mathbf{g}_t^2] = \mathbf{v}$ for every t , prove that

$$\mathbb{E}[\mathbf{v}_t] = (1 - \beta_2^t) \mathbf{v}.$$

(c) (3 pts) With $\beta_1 = 0.9$, $\beta_2 = 0.999$, numerically evaluate the factors $1 - \beta_1^t$ and $1 - \beta_2^t$ for $t \in \{1, 5, 50, 500\}$ and comment on the regime where the correction *stops mattering*.

(d) (3 pts) Compare Adam with **SGD + Momentum**. Give *one* situation where SGD+Momentum is preferable, and *one* where Adam is. Justify both in terms of loss-surface geometry or data structure.

End of Homework 1 – Theory Part